

Module 03: Perception in Crochet Circles

Learning Objectives

1. Explain how **yarn tension** provides continuous proprioceptive feedback — a sensory channel through which the crocheter perceives the state of the fabric.
2. Describe how **dropped stitch detection** works as a surprise signal — a prediction error that alerts the crocheter to an unexpected change.
3. Identify how crocheters discriminate between **yarn textures and weights** by touch, and relate this to precision-weighted sensory processing.

Introduction

Close your eyes and crochet a few stitches. You will find that you can continue working — at least for simple, familiar stitches — without looking at your hands. How? Because crochet is not only a visual craft; it is a deeply tactile one. Your fingers sense the tension of the yarn, the resistance of the hook passing through a stitch, the texture of the fiber, and the weight of the growing fabric. In Active Inference, **perception** is the process by which an agent infers the state of the world from sensory observations. For a crocheter, perception happens as much through the fingertips as through the eyes.

This module explores the sensory world of crochet. We will look at how tension sensing acts as a form of proprioception, how the surprise of a dropped stitch reveals the crocheter's implicit predictions, and how experienced crafters develop remarkably precise tactile discrimination.

Key Concepts

1. Tension Sensing as Proprioception

Proprioception is the sense of the body's own position and state — the feeling of where your limbs are, how much force you are exerting, and how your muscles are engaged. In crochet, proprioception extends through the hook and yarn to encompass the entire work-in-progress.

The yarn running through your tension fingers (typically the index and middle fingers of the non-hook hand) provides a continuous stream of information about the state of the system.

In Active Inference, sensory states at the Markov blanket carry information from the external world to the agent's internal states. In crochet, yarn tension is one of the most important sensory channels. When tension is even and consistent, the crocheter infers that the system is in a predicted state — the yarn is feeding smoothly, the stitches are forming correctly, and the fabric is growing as expected. This is **perceptual inference**: the agent's best guess about the hidden state of the world (the fabric) based on sensory evidence (the tension).

When tension changes unexpectedly — the yarn snags, a stitch is too tight, the yarn runs out — the crocheter immediately notices. This is because the sensory signal deviates from the predicted signal generated by the generative model. The crocheter expects a certain resistance when pulling yarn through a stitch. If that resistance is suddenly higher (a tight stitch) or lower (the yarn has split or the hook has missed the stitch entirely), the prediction error generates a feeling of "something is wrong." This feeling precedes conscious identification of the problem. Many crocheters report that they "felt the mistake" before they understood what happened.

The tension finger acts as a precision instrument. Experienced crocheters maintain remarkably consistent tension across hundreds of stitches, producing even, uniform fabric. This consistency is not mechanical — it is the result of a highly refined perceptual loop in which the crocheter continuously senses tension, compares it to the expected value, and makes micro-adjustments in finger pressure. This is Active Inference in its most elegant form: a tight perception-action loop operating below the level of conscious thought.

2. Dropped Stitch Detection as Surprise

In Active Inference, **surprise** (technically, surprisal) is the negative log probability of an observation under the agent's generative model. High surprise means the observation was unexpected. Low surprise means things are proceeding as predicted.

Detecting a dropped stitch or a missed insertion point in crochet is a vivid example of surprise. The crocheter's generative model predicts a certain rhythm: insert hook, yarn over, pull through, yarn over, pull through two. When a stitch is missed — the hook slides past a stitch

top without catching it, or a loop slips off unnoticed — the rhythm breaks. The count is off. The fabric looks different. The crocheter experiences surprise.

What makes this particularly interesting is how quickly the surprise is detected. An experienced crocheter often catches a dropped stitch within one or two stitches because the tactile feedback changes. The missing stitch leaves a gap that alters the tension on the working row. The hook meets less resistance at the next insertion point, or the fabric has a subtle looseness. These are tiny signals, but the crocheter's finely tuned generative model detects them because it generates precise predictions about what each stitch should feel like.

For beginners, dropped stitch detection is slower. Their generative model is coarser — it predicts that crochet should feel "roughly like this" but cannot distinguish between the feel of a correct stitch and a missed one. As they gain experience, their model sharpens. They can distinguish between accidentally working into only one loop (instead of both) of the previous stitch, inserting the hook in the wrong space, and completely missing a stitch. Each of these errors produces a slightly different sensory signature, and the experienced crocheter's model can tell them apart.

3. Yarn Texture Discrimination and Precision Weighting

Active Inference uses the concept of **precision** to describe the reliability of sensory signals. High-precision signals are trusted more by the agent; low-precision signals are weighted less. In crochet, the crocheter's ability to discriminate between different yarns by touch — and to weight that tactile information appropriately — is a clear example of precision weighting.

An experienced crocheter can pick up a strand of yarn and tell you a remarkable amount about it from touch alone: it is cotton (smooth, slightly stiff, with no bounce), or wool (springy, with a slight halo), or acrylic (smooth, slippery, with a plastic feel). They can estimate the weight category (fingering, sport, worsted, bulky) by the thickness between their fingers. They can feel whether a yarn is plied or single, tightly spun or loosely spun. This is the result of a generative model that has been trained over years of handling different fibers.

In Active Inference terms, the crocheter assigns high precision to their tactile sense when evaluating yarn. They trust their fingers. But this precision is context-dependent. When working in dim light, tactile precision increases because visual information becomes

unreliable — the crocheter weights their proprioceptive and tactile channels more heavily. When working with an unfamiliar fiber for the first time, tactile precision drops because the generative model does not yet have strong predictions about what this yarn should feel like. The crocheter works more slowly and pays more deliberate attention, giving the model time to calibrate.

This is also why many crocheters have strong preferences about yarn: "I love the hand of this merino" or "I hate working with slippery nylon." These preferences reflect the crocheter's perceptual system — some yarns produce sensory signals that are easy for the generative model to predict and process (high precision, low free energy), while others produce noisy, unfamiliar signals that require constant recalibration (low precision, higher free energy).

Applications

In crochet, we see Perception manifest in:

- **The Eyes-Closed Test:** Many experienced crocheters can continue simple stitches with their eyes closed, relying entirely on proprioceptive feedback. This demonstrates that the tactile sensory channel is sufficient for perceptual inference in well-learned tasks. The generative model generates predictions about what each stitch should feel like, and the hands confirm or deny those predictions without visual input. Trying this exercise with a familiar stitch versus an unfamiliar one reveals exactly where the generative model's precision breaks down.
- **Stitch Counting by Feel:** Some crocheters do not count stitches by looking — they count by the rhythm of the hook. Each stitch has a tactile signature (insert, yarn over, pull through), and the crocheter can count repetitions of this signature without visually inspecting each stitch. When the count feels wrong — a row ends too soon or too late — the crocheter detects the error through a mismatch between the expected rhythm and the observed one. This is perceptual inference operating through a temporal, rhythmic modality rather than a spatial, visual one.

Conclusion

Perception in crochet is rich, multi-modal, and deeply embodied. Through tension sensing, the crocheter maintains a continuous proprioceptive loop that monitors the state of the fabric.

Through surprise signals, dropped stitches and errors are detected — often before the crocheter consciously understands what went wrong. Through precision weighting, experienced crafters learn to trust their fingers, discriminate between fibers, and adjust their sensory reliance based on context. In the next module, we move from sensing to thinking: how does the crocheter's mind process all of this sensory information into plans, predictions, and patterns?